Day 5. 4. APIM Part (3)

* Yesterday → Simple demo → how APIM will call
* API proxy → Cpl Syntax
* let us see what are all available options in that particular API proxy
* once you Build this API proxy → policies → Important part
  → Multiple set of policies
  ① Security policies
  ② Traffic management policies
  ③ Mediation Policies.
  ④ Extention policies
* NOW the point is 'how you use' these Policies is nothing But 'API management'

* cpi → Pallets Similarly → API → Policies

* All policies are not required but some of the Policies we will use max (3 or 4) on regular basis

① Traffic management Policies

→ Access Control (Policy)

ip Access Control policy

* How to add this policy

→ Policy Editor

Flows: Proxy Endpoint

→ Pre Flow

→ Post Flow

Target Endpoint

→ pre Flow

→ post Flow

* Proxy Endpoint → Source → API proxy

Pre Flow:

* Proxy Endpoint called → Before to this

→ operations → pre flow

Port Flow : After proxy Endpoint Called

* Target Endpoint

demo : post man → Endpoint URL (API proxy URL)

→ Post Method → Result

Without Authorization
we will get result !!

The reason is due to APIM !!

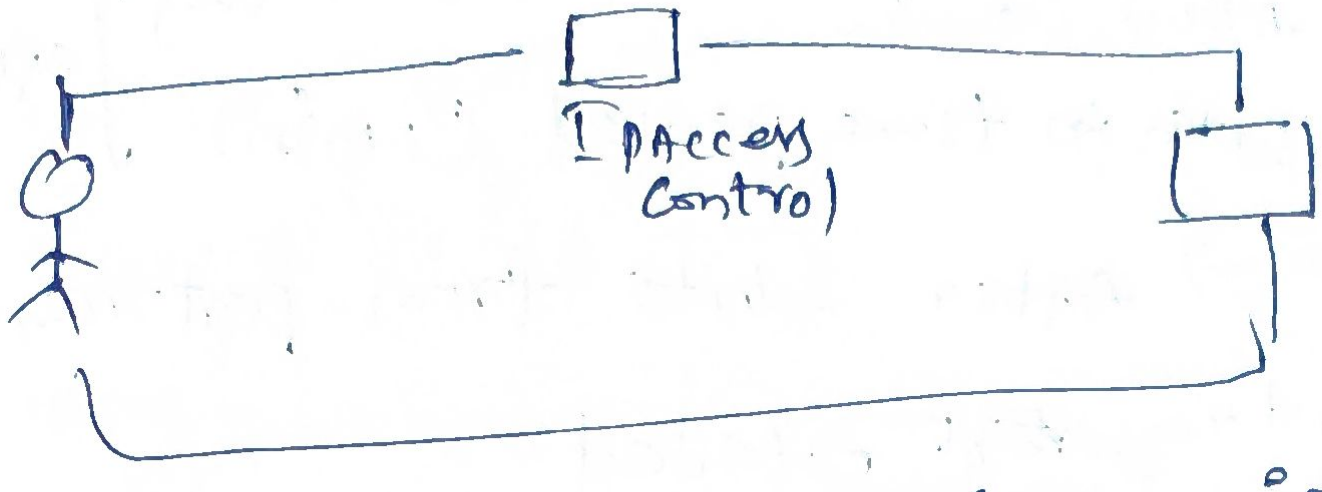
So APIM, if we want to control →
Security Policies → Before → API proxy
call → we need to perform

(*) Traffic management policy

→ Access control (ip Address restriction)

Scenario: if we to restric the request from particular ip address we will use Access control policy

(1) Create policy → [Access Control]



→ If you want get your System ip address first make changes in configuration changes should be → changing ~~to~~ from ~~Allow~~ → Deny (noRuleMatchAction)

→ captured ip in postman

→ MatchRule ✓ (Remember)
↳ "Allow"

* multiple ip's → Duplicate

(2) [Quota] → Policy

* APIM → inbound → S4HANA

* Backend System → 10,000 --- requests
→ not manageable

* Quota limit (Allow, TimeLimit tags are important)
  - Allow Count → 200 reqs } example
  - with in time period (1 min)
* demo → After 2 hits from postman →
  3rd request → failed

→ Quota limit = To control the number of requests in a specified ~~limit~~ time then we will use Quota limit Policy

* when ever we open the policy there will be the default Content
* Default content will be there for each and every policy in SAP doc's website (Blog's)

③ Spike arrest: It is also a one type of Quota limit policy

→ spike arrest → for each and ever request
→ you should have time gap

eg: 10 sec → 2 req → 1st request — 2nd request
2 (sec) — 5 sec

④ Reset Quota : what ever the quota we had already set that limit we can reset using Reset quota policy

eg: 10 pm

* If we want reset the quota limit for particular user then we will use reset quota

* If we want to reset the quota limit based up on the apikey then we will use Reset quota

query param

→ request.header.apikey

header

key value

apikey

* Scenario → particular Customes → reset quota limit

identification of customers

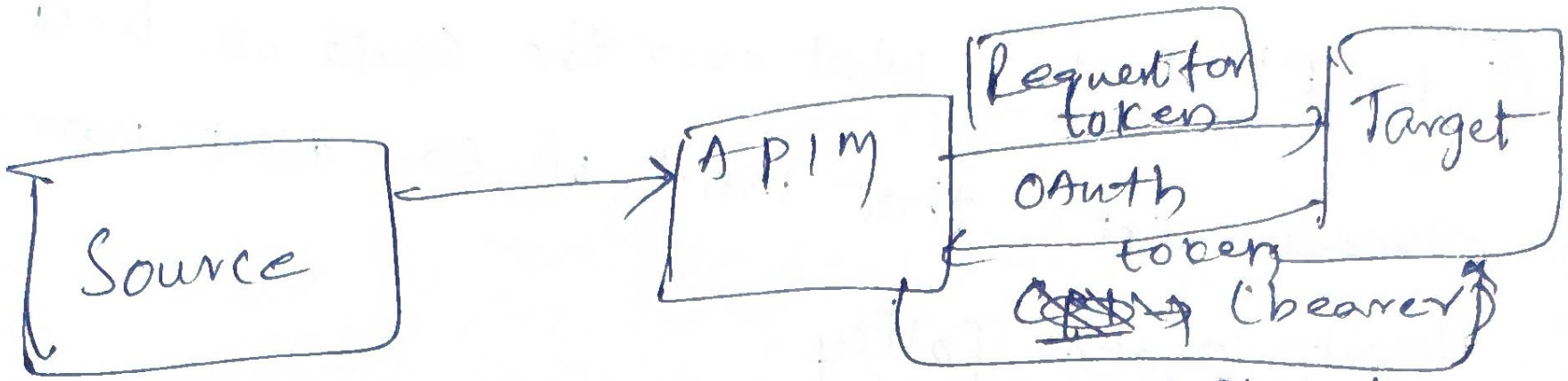
⑤ Cache policy * data is stored temperorily

frequent files

CPI → Security materials

APIM → ?

} for OAuth Authentication



* here in CPI perspective → Typ OAuth credentials
  → in CPI → monitor → Security materials

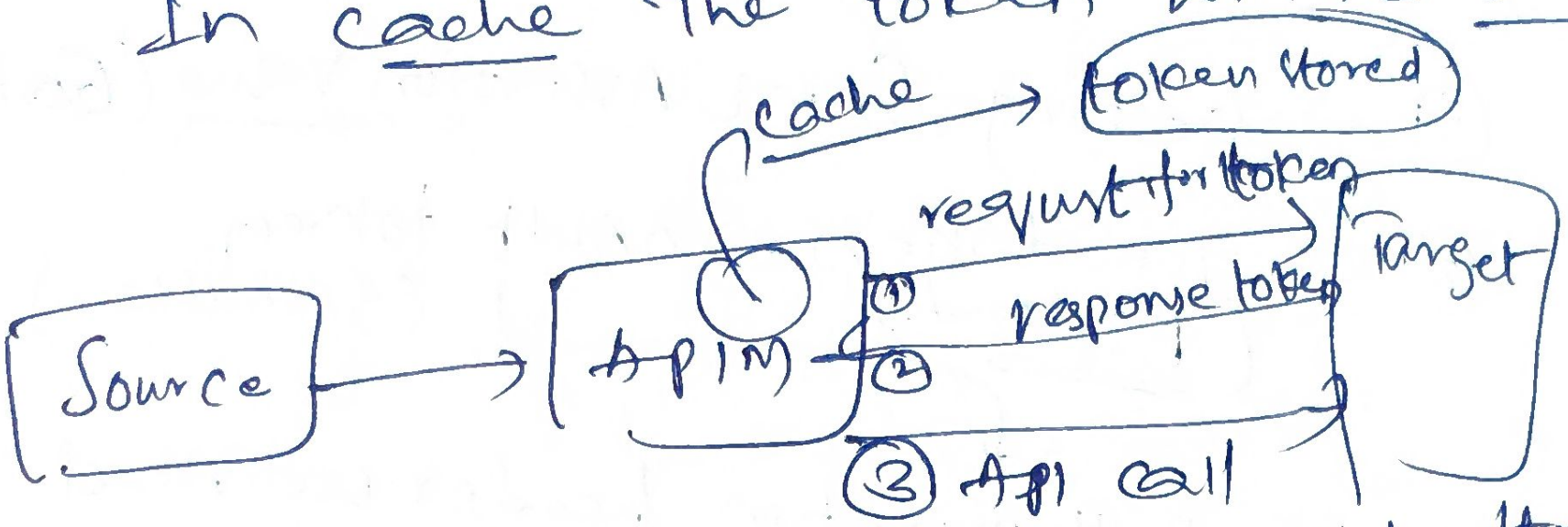
But in APIM

* In APIM we don't have Security materials or like that then what we will do

Scenario: 1day active token → 200 requests →
200 times request for token → Target
Request → Time waste

Soution: APIM → Cache → Token → Stored
first call (1st request)
2nd call → APIM → Cache → gets to Key
→ then Target

* So here the main important point is In cache the token will be stored



* from 2nd request the token will be fetched from cache

* Standard Content → Discover Tab

* Discover → APIs → already developed by SAP

demo:

SuccessFactors_OAuth2SAMLBearerAssertion

Copied from pre built SAP API's

→ Configure → APIs → Policy Templates.

* If you want to Connect to the SuccessFactors from SAP CPI APIM We weneed to make two API calls

* 2 API calls → fetch token. (getSAMLBearer)

① → Getting SAML Assertion Value (Grant)

② → get token → OAuth token (getOAuthToken).
(payload)

Authorization header we need to pay

* popular cache
* lookup cache

* So here the question is we need demo on getting the OAuth token from any target and storing in Cache and using that token in APIM

∴ for the above task we need to do configuration, how to do that?